

Human-centered remanufacturing scheduling

Collaborative learning modes, evolving skills and cognitive workload

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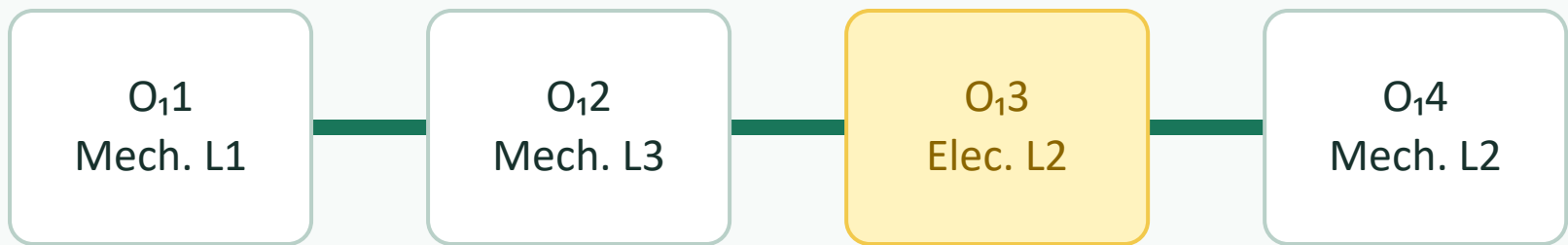
Why this problem?

Remanufacturing keeps products in use—but its uncertain operations still depend on skilled people.

- Workers have heterogeneous and evolving skills
- Training can occur during production
- Collaboration changes time and cognitive effort

What must be scheduled?

JOB J_1



Example: O_{13} requires electrical skill level 2

Worker eligibility

W1	Electrical L1	Teaching
W2	Electrical L2	Solo
W3	Electrical L1	Teaching

The schedule decides WHEN, BY WHOM and IN WHICH MODE each operation is performed.

Execution modes

SOLO
Qualified worker performs alone

TEACHING
Expert + learner; skill improves

COLLABORATION
Two workers; faster, more workload

EXTENSION · SOLO WITHOUT REQUIRED SKILL
Allowed only for a one-level gap, with time and quality penalties.

Lexicographic objectives

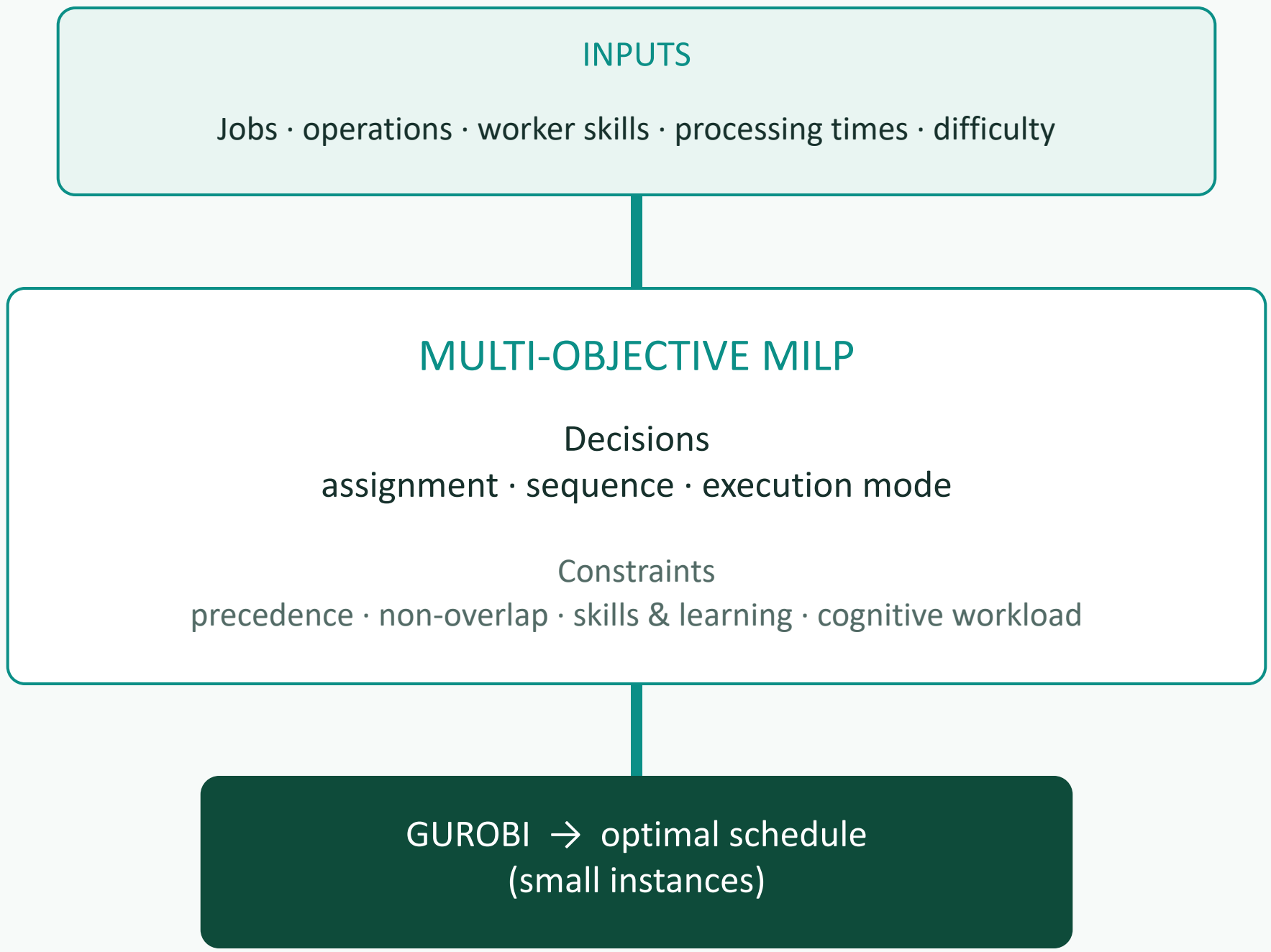
Priority is strict: ① benefit → ② skills → ③ cognitive workload

1 MAX BENEFIT
 $\sum_i w_i \cdot \text{job_done}_i + \sum_{ij} \text{task_done}_{ij}$

2 MAX SKILL DEVELOPMENT
 $\sum_k \sum_m \text{skills}_{km}$

3 MIN COGNITIVE WORKLOAD
 $\sum_k \sum_m \text{CL}_{km}$

Exact method: MILP

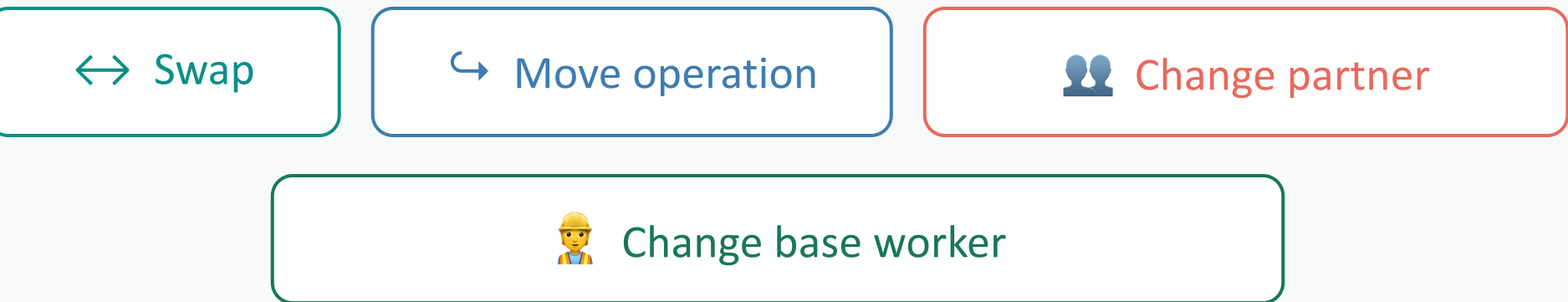


Scalable method

A compact solution encoding links operation order and worker assignments.

OSC	21 11 12 22 13 14
FWAC	2 1 1 2 1 3
SWAC	– – 3 1 – 1
MODE	A A C T A T
BWAC	1 2

Neighborhood operators



ONE MODEL · TWO SOLUTION SCALES

MILP validates optimality on small instances; the metaheuristic targets larger instances.

Results

Reserve the largest visual area for evidence—not additional model detail.

① Representative schedule

Large Gantt: color Solo / Teaching / Collaboration

INSERT FINAL FIGURE

② Human-factor trade-offs

Benefit vs. skills vs. cognitive workload

INSERT FINAL FIGURE

③ MILP vs. metaheuristic

Solution quality and runtime by instance size

INSERT FINAL FIGURE

Key takeaways

- Quantify the value of learning and collaboration.
- Show when cognitive workload changes the schedule.
- Report the scale at which the metaheuristic becomes necessary.

Replace these prompts with 2–3 quantitative conclusions once experiments are final.

QR
code

PROJECT RECIRCLE
Human-centered optimization for remanufacturing
Contact: Yassin Lahbib · Add email / repository link

MAIN ASSUMPTIONS

Workers remain available over the planning horizon. Teaching and unqualified solo execution are limited to a one-level skill gap.